PART 2

Port Cities

CHAPTER 3

Port Cities as Urban Assemblages. Bringing Actor-Network Theory to Maritime Sociology

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1 Introduction

The cities of the 21st century have become the objects of cognitive experiments within the expanding urban studies (Robinson, 2015). These experiments were accompanied by criticism of the paradigms being decreasingly useful in the explanation of the dynamics of changes in contemporary cities. Some studies were criticised for its political agenda rather than the observation of urban reality (Sapotichne et al., 2007), others for being rooted in the professionalised routines and outdated (Brenner et al., 2011), overfilled by contrary interpretations. Other critics complain about the Anglo-American hegemony in urban studies (Robinson, 2015).

The multiplicity of experiences burst with new conceptualisations and interdisciplinary research methods but also reflect positively on the inclusion of the natural environment as an important urban factor. In recent urban studies, the complexity and dynamics of networked urban systems have been extensively reinterpreted with Bruno Latour's (2002, 2005) Actor Network Theory (ANT). Rooted in the sociology of translation, urban studies focus on materiality, artefacts and objects by implementing ANT in such domains as: (1) identifying missing, obscured or invisible non-human actors in housing, market and policy practices (Gabriel & Jacobs, 2008), (2) interpreting processes and mechanisms in order to distinguish active mediators and passive intermediaries (Cowan et al., 2009), (3) developing interpretive tools for complex associations of people and objects in architecture, spatial planning and governance (Fallan, 2011; Müller, 2015) and (4) assessing the aspect of urban assemblages for planning and building sustainable cities (Cvetinovic et al., 2017).

Following these research achievements, we recall the argument that providing the conceptual framework within urban studies is necessary and possible by combining the concepts of urban assemblages and ANT (Latour, 2002, 2005; Callon, 1984; Law, 1987). We regard it useful especially for the interpretation of port city transformations and the network of actors responsible for those changes, exploring forms of relations, and identifying of port-city actors

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[nature (sea), technology (port) and culture (local maritime communities)]. Even though ANT is criticised both as a theory or methodological principle (Radder 1996), there is an increasingly evident viewpoint that The Actor-Network Theory and its post-development might change urban studies and the way how we perceive the city. We intent to point out that a conceptualisation of the city-port relationship in the ANT perspective will result in a revision of the theory of urban studies and maritime sociology with more accuracy in the analysis of the links between actors and the environment. As a conclusion but also a starting point for further work, we propose a research agenda inspired by this post-humanistic approach [bearing in mind its criticisms (Brenner et al., 2011)].

2 The Port and the City. In Search of a Common Perspective

Port cities are complex social organisms. Despite their long history and connection with the processes of civilisation and urbanisation, their unique character still escapes theoretical reflection within the framework of urban studies. The difficulty in describing a port city is related to the globalisation and transformation in both the city and the port itself as increasingly autonomous, automated and multi-modal. Most of the "classical" concepts and notions of the port city referred to their selective aspects, narrowing the phenomenon to historical, functional, cultural, economic or spatial analysis (among others: maritime industrial development areas (Vigarie, 1981), maritime economy (Hoyle, 2000), port region concept (Ducruet, 2009), the gate city (Bird, 1983), the economic cluster (De Langen, 2004), the growth pole (Fujita & Mori, 1996) and the port city as a growth machine (Jaffee, 2015)).

Recent studies have shown that the global economy is creating a new context for port cities (Ducruet & Lee, 2006). The global network of port cities has become a reality which has led to a new way of organising maritime affairs (Ducruet, 2016). To make things a little simpler, the two seemingly contradictory phenomena are reflected in the port city concepts:

1. the port as a collective actor distancing from the city,
2. the port as part of a complex transport and logistics network, embedded within multi-scalar regimes, integrated with inland regional infrastructure.

They do not contradict each other, as the ports, influenced by the need to increase their own handling efficiency, are forced to use the inland area to maintain their communication with global transport and logistics systems.

The perspective of a port, therefore, goes beyond the city which is guided by different development priorities which often contradict the former.

In the first case, urban studies describe a spatial and functional separation between cities and ports. Ports are moved outside the city boundaries which is caused by the need for the spatial development of the city and the port (and its needs) itself (Monios et al.,2018; Wiegmans & Louw, 2011). Competition for space between port environment entities and other urban actors not related to the maritime sector is gradually pushing port areas out of the city area. In the second case, the port regionalisation concept suggests that ports are competing to serve overlapping hinterlands, intermodal connections and suitable logistics structures. The port's transition between local, regional and global scale results in creating multifaceted port gateway regions (Monios & Wilmsmeier, 2013).

This reduces the dependence of ports on the urban labour force, which only deepens the process (Hein & Schubert, 2020). Inland port cities are subject to the lock-in effects of metropolitan cities (Witte et al., 2019). Other ports specialise in the narrow field of transport, limiting themselves to service functions towards central metropolitan ports. The question arises whether port cities lose their specificity. How is the image of the city-port relationship changing?

It should also be stressed that the port city is a hub of social and cultural exchange located at the sea-land border. In other words, it is a centre where the movement of people, goods, services and ideas through maritime links is concentrated (Kokot, 2008; Hein, 2013; Mah, 2014). What is important, in this case, is the existence of the added value of the port which acts as a centre of influence on the social environment through the creation of maritime and port clusters, attracting capital and creating jobs. Integrated knowledge transfer hubs are built around port cities in areas such as management, the transport system, mobility policy, tourism development policy, territorial cooperation, etc. A port in the city is becoming an autonomous, independent enterprise, separated from cities and its inhabitants, and continues to grow in order to serve more vessels and more transshipments (Grossmann, 2008). Rationalisation and the power of technology dominate over romanticism: the ship and its place of loading have become inaccessible, port districts have disappeared in many cities and the port no longer serves information purposes. The emergence of new technical solutions, such as ships with limited crews or automated oil and container terminals, has in many cases resulted in the dehumanisation of work at sea (Kahveci, 1999). The port becomes a self-sufficient actor who undertakes its activities in isolation from everyday life in the city (Hall & Jacobs, 2012). On the other hand, the port regionalisation concept

suggests that the port is not a passive actor but plays a role of the collective creation of ties and relations between itself and the inland actors. In fact, the port authority should act as a facilitator of relations between various stakeholders, i.e., real estate developers, port terminal operators, shippers, carriers, labour and government bodies (Notteboom & Rodrigue, 2005).

Considering the port city only from a transport and economic point of view loses an important aspect of the exchange of culture, knowledge and information. Modern research on the port city is becoming 'too much economical', in terms of prospects dedicated to improving transshipment rates and increasing the efficiency of port space use. The influence of the sea and the port facility on the social awareness, institutional order and quality of life of its inhabitants, and actions taken in the context of the sea, is underestimated.

The sociological perspective sees the city through the prism of aesthetic and social diversity, as a metaphorical area saturated with myths, stereotypes and symbols. Port cities are primarily local communities that exist in a specific space, linked by forms of collective life, and in particular social institutions that, through their proximity to the sea, take on specific professional and sociocultural activities. Port cities are on the verge between 'black and blue', as Mah (2014) has described it, for centuries as exotic places of cosmopolitanism and a vibrant cultural exchange associated with the "blue" sea. These cities are to be filled with blue, which is the colour of longing (Mentz, 2009), imagination and challenges and black as "dark" places of crime and violence. In this collage of the ambivalent urban identities, port cities reach deep into our culture and consciousness, through ideas, stereotypes, fears and prejudices, creating a specific image of the city. This applies not only to the workers of the maritime economy - seafarers or fishers and their families, which form a different social world (Maurizio, 2013). The proximity of the sea connotes the emergence of various domains of people's lives that are different from those in inland cities. However, our spatial imagination focuses mostly on ports and spaces, such as quays, terminals, ferry crossings, waterfronts, promenades, beaches, marinas, fish markets, lighthouses, piers, port cranes, former granaries, etc. (Kowalewski, 2021).

The multiple influences of the sea are also visible through the institutions and maritime economic activities. They create important links with those working directly at sea. They are not only institutions and organisations of a professional nature (i.e. vessel managers, ship owners, crew manning agencies, ship maintenance or insurance companies) but also those indirectly connected with the sea such as maritime offices, port authorities, shipyards, seafarers' homes, trade unions, naval academies, maritime universities (Kołodziej-Durnaś, 2019). Moreover, professional activity related to the sea influences relationships and

dependencies between professional roles in the maritime economy and family, neighbourly and civic roles. This manifests itself in the following spheres: education (maritime upbringing programs at schools, sailing education and training) (Lau & Ng, 2015; Kołodziej-Durnaś & Kołodziej, 2017), cultural events (festivals of the sea, regattas, shanties) (Gallagher & Pike, 2011), sport and recreation activity (fishing or sailing clubs, local associations), local maritime local traditions (maritime baptism, concerts for 'those who have not returned from the sea', ship launching ceremonies, sailing ship parades) (Ransley, 2011), symbols located in the city space (anchors, masts, monuments), as well as specific features of the professional language of, as part of the maritime culture (Demydenko, 2010).

As a result of interweaving these roles and social relations, a specific type of social life in the city emerges. Ludwik Janiszewski (Janiszewski, 1988; Kołodziej 2015) made one of the first attempts to put these many threads into a coherent scientific theory. His 'theory of marinisation' was an attempt to explain theoretically how the sea affects social phenomena and processes. The influence of the "sea factor" concerns all the maritime elements in the material and non-material culture of society. For Janiszewski, in the most general sense, human activity related to the sea meant a whole range of purposeful activities, differentiated by their direction, form, type and degree of connection with the sea, aimed at satisfying individual and collective needs. It is, therefore, a system of intentional actions expressed in the conscious behaviour of individuals and groups of people. He stated that human activity related to the sea was not limited to economic activity, as the sea was also to be a domain of political, cultural and social activity. Marinisation is a process that has accompanied humanity for centuries, as is industrialisation and urbanisation. Like other processes, it appears with the growing economic importance of large companies and professions associated with the sea. Those activities raise the standard of living by an increased ability to meet needs through the sea. The effects of the marinisation process are multifaceted: spatial (about half of the world's human population lives near the coast), demographic, economic (access to the sea also means access to maritime resources, favourable climate zone for tourism, more opportunities to invest in renewable energies, ports or shipyards). Marinisation theory also suggests that, as a result of human activity connected with the sea, a local maritime community, characterised by rules of interaction, social relations, attitudes, values and norms different from land-based communities emerges. This manifests itself in the following aspects: folk tradition, art, maritime culture or the so-called 'maritime consciousness' (i.e. specific features of personality, mentality and imagination). While the first three aspects are even observable today, it has not yet been proven that there

are separate local maritime communities in port cities or smaller fishing villages. Likewise, it has not been possible to explain how the process of marinisation actually translates into the opinions, attitudes and behavioural patterns of people living under the influences of the port city and the sea. This was to be remedied by Polish maritime sociology, which, as a specific research area, was to prove the need for a new sub-discipline in sociology as research progressed.

At that time, the necessity of embedding social relations within the larger biophysical world also became a subject of interest for environmental sociology and human ecology. As the ecological constraints on social problems, such as limits of growth, began to appear in public discourse since the 1970s, there were calls to reject the purity of disciplines in favour of interdisciplinarity. The need to study the effects of the environment on society was reflected in adopting "a new ecological paradigm" (Catton & Dunlap, 1978). Environmental sociology set a bit different angle by examining how society impacts the environment and what consequences there can be for both. The effect of this discourse was a rash of research on problems related to new social movements, environmental conflicts, reflexive modernisation, risk society, green cities, sustainable development etc. One of the emerging theoretical concepts was social metabolic theory. The theory proposes the explanations that each mode of production generates a distinct social metabolic order that influences the interchange and interpenetration of social and ecological systems. The postulate of the relationship between man and nature followed Karl Marx, who described this relationship as a metabolism, i.e. the process of exchange of matter between human and nature (Lee, 1980). In the spirit of this concept, the professional activity connected with the sea regulates the processes that translate, mediate, and control the metabolism between the environment and society. The course of this metabolic process is supervised by social institutions and economic systems in each social system.

Nevertheless, they are all mutually connected in a world eco-system (Longo & Clark, 2016). In this sense, we ought to discuss the impact of the world ocean as a whole because it is the great regulator, the great stabiliser of temperatures (Carson, 1993). On this basis, John Hannigan (2017) proposed the new special sociological area named "sociology of oceans". The new discipline was supposed to be situated at the interface between environmental sociology and maritime studies.

The third trend of research at the interface between the sea, the port city and the maritime community is the concept of establishing a sociology of the coast. The premise for this idea is the fact that researchers often ignore the coast as a unique space of its own. The dichotomy of sea and land seems artificial as it excludes the fluidity between land and sea. Coasts are spaces defined by a multitude of different practices, formal and informal representations and

the everyday lives of those who live there. From prehistorical ages, coastal places were subjects of people's imagination including gods, supernatural powers, forces of nature, sacred rites and contemplation. Coasts were mystical gateways to interference for both reverence and awe, as well as the deliverer of fear and honor. The end of the romantic approach making space for unprecedented intellectualisation, rationalisation and a loss of cultural myths is, once again, emphasised here. In the context of contemporary culture and the globality of the seas, modern coastal practices turned a rural and sacred coastal space into the post-productivist countryside of consumption. The times of industrialisation and urbanisation, and today post-modernisation, transformed coastal places not only in an economic sense but also culturally and ecologically. The contemporary image of the coasts is oriented around consumption patterns by resting spaces (i.e. idyll beach spaces, promenades, waterfronts, tourism resort lays, open-air ethnographic museums, lifestyle enclaves, gentrifying residential estates), icon landscapes for business and policy campaigns or mass cultural events, sites for contestation over public/private land tenure or climate change. If we want to find an answer to the question of how the sea impacts social awareness, we need to refer to the mysticism hidden in the personal lives of people and their cultural practices. Referring to Max Weber's concept of rationalisation, Nick Osbaldiston (2018) reminds us that people's perceptions of the sea and its influence on their lives depend on traditions, historical experiences, folk beliefs, professional ethos, myths and legends, the axionormative system, etc. All those are subject to cultural rationalisation, alternative to the paradigm of economic rationalisation. It is easier to observe this in small seaside towns or former fishing villages than in port cities. Nevertheless, the coast is not only a place limited to sandy beaches, or rural (not-urban) areas by the sea.

In common sense, a port is a metaphorical area in people's imagination, including places and spaces no longer used for professional activity related to the sea. The maritime myths of the port city can be hidden in institutions indirectly connected with the sea or social practices, including tourism or mass events. Our viewpoint is that the sea regulates the processes that translate, mediate, and control the course of the metabolism between the environment and society. However, it occurs difficult to prove that, because the sea-impact indicators are difficult for researchers to capture in the course of routine interview questionnaires or surveys. In the light of the issues outlined, we would like to propose the inclusion of a post-humanistic theoretical perspective in these reflections, especially by combining the above theories with the Actor-Network Theory and Urban Assemblage concept. We will present the reasons in the next part. Nevertheless, this idea was already initially signalled by Arkadiusz Kołodziej (2019).

3 Port City in an Actor-Network Theory (ANT) Perspective

The main question behind ANT is about what relations or associations cause something to become social. The answer to this question refers to three principles described by Michel Callon (1984): the extended agnosticism, the generalised symmetry and the free associations. The 'social' is a specific type of associations that can be described by activities undertaken by humans and non-humans. The principle of generalised symmetry urges to treat all entities, like natural, technical, institutional and cultural equally as they are all engaged in a hybrid system (Callon, 1984). One should note that ANT involves controversies by opening towards dimensions and agents not considered in urban studies. The definition of the city in ANT's way would depend on the author's stock of knowledge, his/her repertoire of theoretical perspective, research methods and analytical tools. The definition itself would be only one possibility to depict "the city" in actor-network of associations, as it will be decided by factors such as the scope of associations and agents, followed by clarity, consistency of the arguments presented. If we apply Callon's principles to the city, we will see economic, spatial and cultural dimensions of the same object. ANT approach offers the alternative explanation of the city as it is not a product of economic unit or a social construct but a result of associations between networks of actions taken by objects, natures and humans (Law, 2006).

The Latourian method combines what social and 'natural', fuzzing the line between nurture and nature. The Actor-Network can be defined as any collection of actants (human, non-human and "hybrid" human/non-human actors) jointly participating in some organised (and identifiable) collective activity in some fashion for some period of time. In Latours' (2005) proposal "any thing that does modify a state of affairs by making a difference is an actor" (p. 71). The human and non-human actants need not know one another, but they may serve one another as legal partners, resources, competitors, dream figures, reference point etc.

The actors/actants in the network do not act chaotically, as they are motivated by a particularly understood interest. The interest of a human actor is a content-specific motivational characteristic composed of intrinsic feeling-related and value-related valences (Schiefele, 1991). The interest of a non-human actor is the purpose for which it was designed to serve (technology) or the way it was chosen to be used (nature) (Tatnall & Lepa, 2003).

ANT's research approach attempts to analyse a certain set of actions "whereby the respective interests of different human and non-human elements are aligned into a social [...]" (Gao, Damsgaard, 2007, p. 185), technological and natural arrangement (Gao, 2005). Actions of actants mediated by technology

and nature might form an actor-network. However, this is only one possible outcome and one that takes more time and effort than an action net (Czarniawska, 2002). Actor-networks are established when arrangements between actors produce stable patterns of performance and practices (Smith, 2006). In other words, actor-networks are produced and performed in co-presence of objects in locally shared life-worlds (Tironi, 2010). Since co-presented objects are networked entities embedded in action nets, they can produce, absorb, perform, or disseminate a tacit knowledge related to the local context. Tacit knowledge exists in the background of our consciousness and enables us to take actions without the ability to explain how the action is successful in performance. It can be shared effectively between two or more actants when they share a common spatial or social context (Gertler, 2003). Nature as both, non-human actor and part of a hybrid network, translates its activities by technology (non-human mediator) (Tironi, 2010; Guggenheim, 2010). It means that associations of actants are possible through their performances of skills driven by mutual interests, translated by technology and enabled by shared tacit knowledge that happens in spaces distributed throughout the city. In urban research, the entire action nets need, therefore, to be observed in the first place. The action nets must be the starting point, not the result of a study. By contrast, many actor-networks can be revealed from the analysis of the action net. It means that not only the urban perspective but also the macro perspective should be taken into consideration because actants form networks on the basis of their actions that are legally required, enabled or influenced by macro actants or a super-actor.

4 The Port City as an Urban Assemblage

The research made reference to the idea of networks (~~Hein, 2013;~~ Beaven et al., 2016), representations and images of urban-maritime cultures and their identities in global legacies context, especially of waterfront decline and revival (Sepe, 2013; Mah 2014), future of maritime economy in context of smart cities (Ducruet, 2016), and the multifaceted structure of port cities and their global function (Hein, ~~2013~~). However, the use of concepts referring to the relational and post-humanistic perspective is rare (with very few exceptions: Markus Hesse (2018) in his paper recalls the concept of relations and networks to analyse port cities in Europe, using case study methodology and desk research. He takes a relational view for presenting links between the local economy and development policies of the European port-city interfaces. Wang & Sellina (2018) apply the Actor-Network Theory (ANT) as a qualitative analytical framework for examining the possible roles played by multilateral platforms

in facilitating cross-border transport infrastructural projects at regional/continental scale).

The contemporary reflection on the city deconstructs the concept of the "city as a whole" and encourages the search for assembled scenes, actions and actors that constitute networks. If the effect of ANT is to dissolve the city by revealing the networks, the concept of assemblages allows to synthesise the city again – as James Donald (1992) wrote:

> [...] there is no such thing as the city. Rather, the city designates the space produced by interaction of historically and geographically specific institutions, social relations of production and reproduction, practices of government, forms of media communication and so forth. By calling this diversity 'the city', we describe to it a coherence or integrity. 'The city', then is above all a representation. [...] I would argue that the city constitutes an 'imagined community'. (p. 422)

The city as a synthesis exists primarily in the imagination of its inhabitants who retain the ability to incorporate individual elements into the overall picture, into a common arrangement. As Thomas Bender stated (2010),

> The assemblages that constitute the city must have enough interconnection for the imagination to be able to make a whole out of them, even though they are not solidly so. City dwellers can patch together their city. Their imaginations can fill in the spaces, make the connections, and make it a fit place into which they can insert themselves. (p. 318)

In this sense, the urban assemblage is a complex whole, encompassing not only communities and institutions but also facilities, material objects, technology, and nature. We define this complex as action nets of natural, technological, and social actors present in many spheres like, for instance, maritime economy, maritime culture, or maritime education. From this perspective, we assume that a port city is more than just an economic entity or a functional area, as it is the net of actors (human and non-human) defined by their actions in the described spheres. Thus, the port city is a dynamic complex urban system where urban elements (human and non-human actors) being part of networks participate in their conceptualisation and transformation. However, assemblages do not form wholes or totalities. Instead, they designate the processes through which the city comes together through associations between human and non-human elements (Farias, 2010, p. 15): the city is an assemblage that consists of a plenitude of assemblages.

5 Research Agenda

The presented approach is appropriate for examining entities comprising social, technological and natural elements, such as port cities. It is conceptually useful in understanding the complexity of reality and the active roles of natural (i.e. sea) and technological (i.e. harbour/port) actants in shaping urban assemblages. Assemblage refers to the process of arranging and organising various heterogeneous actants, like human/non-human, organic/inorganic and technical/natural. Thus, assemblage theory addresses the city as a multiplicity of actions and interactions that form an ongoing process of conceptualising the city (i.e. the port city).

6 Possible Research Questions and Hypotheses

Q: How are the urban assemblages of the port cities created by action nets of natural, technological and social actors?

H: Urban assemblages of port cities are created by human and non-human actor-networks.

The complexity of the networked urban system, through the prism of Latour's ANT, is perceived as an urban assemblage. The urban assemblage is the complex of action nets of natural (environment, implications of the proximity of the sea), technological (port) and social (local community practices and actions) actors (actants). The main research question is how it is done. All of the human and non-human actors together form a diverse range of associations that contributes a stable set of actions, interactions and translations driven by their aim (interest) under a common context. We assume that in every port city, there are actors operating in the field of education, economy, culture, social policy, security where the sense of their activities (or at least part of them) is directly or indirectly related to the sea. This relationship takes place directly or indirectly through technology (possibly port) that acts as a mediator between the sea and the society.

Q1: What is the role and function of non-human actors (sea, coastal environment) in the network of actors involved in the port-city relation?

H1: The sea is a super-actor whose presence modifies practices and network relationships of actors in port cities.

Referring to the definition of non-human agents provided by Jones & Cloke (2008, p. 80), we assume that, although the sea is shaped, protected and marked

by communities and individuals, it itself remains to some extent independent, changing the nature of social relations, geography as well as economic and political conditions. The sea works outside the categories of purpose and intentionality, and its impact - although changing in relation to human activity - cannot be fully controlled, "setting the conditions" for its users through a variety of characteristics such as depth, the abundance of flora and fauna, degree of salinity or temperature. Here, we refer, among other things, to Jessica Lehman (2013), who points out that the ocean is not only a scene for human activity but also working, transforming, cooperating and behaving in a way that is not always predictable. The recognition that the sea is an actor is a particularly productive way of understanding the intersection of different manifestations of the port city.

Q2: How is the relationship between nature (sea), technology (port) and culture (local maritime communities) in port cities created and maintained?

H2: The port is a technology that serves as the dominant mediator between human (local maritime community) and non-human (sea, maritime) actors taking mutual actions in many (sub-)actor-networks, particularly in education, economy and culture.

The basic category of ANT is the translation network known as the actor-network. An actor-network is a metaphor. On the one hand, it is an actor connecting heterogeneous - animated and inanimate - actors with each other. On the other hand, it is a network allowing the exchange or transformation of these actors (translations). The concept of translation is important here: the actors act and thus create new networks, thus transforming reality and themselves by including different elements in the network that create hybrids. Nature, as both, non-human actor and network, translates its activities by technology (non-human mediator). The hypothesis is that the port is both, the actor and the actor-network that translate associations between nature (sea) and society (local maritime community) creating new hybrids like maritime education, maritime economy, maritime culture. To speak non-ANT language and at the same time to make things simpler: the activity of the sea disseminated by technology (port) has social significance as long as it changes the shape of our social life. We define the port city as an urban space inhabited permanently by an urban community that has been living in an environment with the mutual impact of the sea. The developmental trajectory of this community is determined by different interest coalitions connected in various ways and

intensity with port facilities, maritime institutions, maritime organisations, social practices and maritime traditions (Bartłomiejski, 2016). On this basis, we assume that we can trace the links between sea and society, in the areas of education, economy and culture, as they are the defining elements of a port city. The black-box concept of Latour (1987) requires the port to be perceived as a complex network of relations between various actors (environment, technology, infrastructure, institutions, communities). These relations are formed by sets of action programmes whose significance and relation remain beyond the awareness of the inhabitants of port cities. Nevertheless, the social communities of port cities cognitively simplify their perception by calling the network of relations "port". In fact, the port is divided into an internal port (e.g. a base, transshipment terminals), an external port (jetties, wharves, specialised infrastructure for handling specific cargoes), a harbour master's office, waterways, storage yards, shipyards, etc. The ports are perceived as integral wholes. In fact, however, they are the node of a network of relations of many active entities, not only local but also global. With this in mind, the questionnaire-based interviewing technique is intended to investigate the port as a technology that serves as the dominant mediator between human (local maritime community) and non-human (sea, maritime) actors taking mutual actions in many (sub-) actor-networks.

Q3: What is the impact of the macrostructural (economic and political) context on urban assemblages of port cities?

H3: The urban assemblages are revealed in a specific actor-networks, where economic and political contexts (macrostructures) translate the relationship between the basic meaning of the term "port city" and its meaning in concrete actions undertaken in different configurations of actor-networks (education, economy, culture).

ANT does not recognise the distinction between micro- and macro-social. There are macro-actors and micro-actors, but the difference between them is brought about by power relations and the constructions of networks (context). Since the actors are all those who act, they are all "of the same size". This means that with the use of the methodological formula "follow the actors", one can move from the human being to the big, international corporation, organisation or state government, to technology (i.e. the port), and then to natural phenomena (i.e. the sea). ANT does not define which actors it considers to be the main ones, and which actors it considers the background. This would only happen if it gave priority to a certain kind of bonds (i.e. the impact of the sea in the

port city context). Therefore, ANT, on the one hand, like objectivist trends, has no problem accepting the existence of "great" actors. On the other hand, like interactionalist approaches, it asks in which way the larger actors emerge in the negotiation processes (translations). Latour states that larger actors are the result of stabilising relations between smaller actors. The context of the actor-network of the city is based on regional, national and global conditions. These conditions are also actor-networks and should therefore be taken into account in the analysis of urban assemblages of port cities. Krzysztof Abriszewski (2012) argues that it is not worth, when researching networking processes, to a priori limit attributes of actors according to a selected social theory, because each one of them is going to concentrate on one type of relationship with the use of a selected metaphor (e.g. conflict, harmony and organism, negotiations, processes of marinisation, etc.), forgetting about others. Selecting a theory for ANT analysis will create a contradiction in that case. The researcher will empirically examine how the relationships between the actors have been established and disintegrated, describing themselves. On the other hand, by the power of the chosen theory, he will introduce a certain "super-actor" to the game, by whose presence he will explain everything. Such a super-actor may be, for example, the spirit of capitalism, Western rationality or the impact of the sea. To introduce a super-actor (sea) means to try to reduce the explanation to escape from an empirical description to indicate one basic thread binding everyone. We would like to point out that it is not our intention to test Ludwik Janiszewski's theory of marinisation. He took the point of view that the sea affected society and called this process "marinisation". In a similar way, the process of industrialisation affected the economy and the process of urbanisation affected the process of concentration of the population. We know that port cities are created and transformed ("conceptualised") by the social practices of their inhabitants in a specific context relating to the sea and the port. By exploring the assemblages of port cities, we would like to explore the relationship between the basic meaning of the term "port city" and its meaning in concrete actions undertaken in different configurations of actors-networks (education, economy, culture). We do not forget, however, that the city itself can also be perceived as an actor-network, thus modifying its action in relation to other actors-networks (macrostructures, i.e. state and local government, corporations, global economy).

7 Discussion and Limitations of the Proposed Concept

There are several groups of arguments critical of Latour's theory. Some of them concern the secondary nature of ANT in the social sciences (in fact, urban

TABLE 3.1 Proposed research framework

Stage I. Desk Research
Aim. Identification and analysis of key features of port cities – focusing on how port cities translate their sub-programs of activity under the influence of sea relations and who is the dominant actor in building these relations, in each of the examined cases. **Data.** Quantitative and qualitative (Public statistic data, strategic documents, research reports, articles published in local and regional press, brochures, leaflets, posters, partnership agreements, letters of intent, projects. Sources: public statistics institutions; private entities (recruiting agencies, maritime training services, yachting training centres); port authorities and institutions providing services; higher education system (maritime and technical universities), schools and institutions supporting youth education, public institutions, non-governmental organisations, sponsors, organisers and partners of maritime events, Fisheries Local Action Groups, yachting clubs **Methods of analysis.** Descriptive statistics: indicators of maritime economy and port functions (international maritime traffic and passenger flows in seaports, maritime passenger transport and the coastal transport fleet, turnover of cargo in ports, containers in international maritime traffic, cargo vessels entering maritime ports by their flag, fishing vessels by homeport, catches of marine fish by species); local labour market (the structure of employment by economic sector, social and occupational structure); Image of port cities (maritime culture in the local tradition, maritime education programs, sailing education initiatives, memorial events, the activity of local entities)
Stage II. Researching individual and institutional actors
Aim. Interviews have two objectives: 1) to reconstruct the process of creating a network of relations between cities, ports and the environment 2) to identify opinions and knowledge about the role of public and private actors - institutions representing the infrastructure of the port city community. Expert interviews will be aimed at deepening the knowledge on the quality of relations between actors from port cities, formal and informal (two-, three-sided) and networks of (multilateral) relations. **Data.** Quantitative and qualitative data (PAPI/CATI/CAWI survey, IDI, FGI). The selection of the sample should be based on the analysis of secondary data, referring to social actors who may potentially have a direct or indirect relationship with the sea (including, but not limited to: Local Fisheries Groups, maritime clusters, maritime offices, consortia of employers, maritime training agencies, maritime and technical universities, trade unions, labour market institutions, local and regional press for maritime issues, health service, cultural institutions, schools and educational institutions, shipyards, ship owners, shipping agencies, employment agencies, maritime forwarding companies, other entities of the maritime industry).

TABLE 3.1 Proposed research framework (*cont.*)

Methods of analysis. (1) Quantitative analysis. Variables included: roles (affiliation (organisation/sector), connections and collaboration), relation (leadership, partnership, opposition), interests, goals, relations, resources being the basis for cooperation (knowledge, access to information, resources, infrastructure), activities (meetings, joint implementation of projects), distance. Responses will be statistically analysed including measurement of interactions between the actors of the emerging network at the level of institution and the environment, classification of the profile of the examined institution (regardless of the formal sector of activity). The outcome will be transformed into a data matrix for Social Network Analysis stage.
Stage III. Modelling a network of actors using the Social Network Analysis (SNA)
Aim. Identification of the presence of individual actors on the basis of visual network diagrams - identification of distances, nodal points, points outside the network. The analysis of the network, based on interviews, will make it possible to see the complexity of the processes taking place between the actors (companies, institutions, organisations) The analysis of actor-networks will provide an insight into stability, intensity, symmetry and the reasons for relations. **Data.** In order to apply ANT principles for the analysis of urban assemblages we propose to use step-by-step methodology: 1) to identify human and non-human actants and their networks (including set of responsibilities, levels of decision making, formal and de-facto interests, the context of actor-networks), 2) to track the stability of associations among established actors (who translate actions? – what technologies are mediators?), 3) establish the centrality of actors (what is the presence of actors in actor-networks decided from their number of relations, types of relations, the notion of enactment, the functionality of their actions), 4) to search for possible super-actor (what actor is intermediating most of the actions as well as the most meaningful in sub actor-networks). **Methods of analysis.** (1) Visual analysis - using ~~Cytoscape~~ software for visual mapping to develop a network model (connections: nodes, paths), distribution (density, centrality, closeness, distance degree), clustering/segmentation (subnetworks). (2) Statistical analysis: the relationship between the number of relations (degree) and the number of institutions, regression models, clustering/similarities analysis, stochastic blocks models.

assemblages can be treated as a modified theory of the urban system). As Afeltowicz, Pietrowicz and Sojak (2020) write:

> The introduction to ANT's assumptions is extremely simple: in fact, it is a very poor conceptual system (actor/actant, relation, translation, black box,

> stabilization) containing several ontological assumptions that are inconsistent with common imagination (anti-personality, relativity, processability). Meanwhile, Latour not only does not lecture ANT clearly. He is also able to change terminology completely from book to book. (p. 22)

According to Radder (1996), the problem with ANT theory is not so much about its banality, but the repetitiveness (replication) and universality of research using ANT.

Storper & Scott (2016) draw attention to the excessive eclecticism, apparent universalism and the descriptive and apolitical nature of ANT theory and the concept of urban assemblage. Other authors, such as Dragon Simandan (2017), point out the unclear relationship between contingency and necessity in this theory. According to Simandan, both forms of inference should be treated as a continuum, and in the theory of urban assemblage itself, it is possible to determine the range of possible variants/configurations resulting from external factors.

Another group of arguments no longer concerns ontological and epistemological assumptions but methodological issues, such as, for example, the problem of decoupling the categories of network actors. As far as urban assemblages are concerned, we believe that this problem can be minimised by comparing both the elements of the assemblage and the relationship between them. If we assume that ANT has a different ontology, the possibility of comparing results is given by "traditional" methodology. We claim that some problems with ANT theory can also be overcome at the level of analysis by defining research units from "different social orders" (micro- and macrosocial, institutions, people, organisations, texts, architecture, nature, technology, culture etc.). Referring to the data, we can determine how a complex organisation (consisting of people, positions, technology, creating an organisational culture) acts as an actor/actant and their group.

In light of the above criticism, we would like to remind the reader about the need for a change in urban studies we raised in the introduction. We see a chance to overcome the difficulties, raised by the theory of marinisation (maritime sociology), social metabolic theory (ocean sociology) and sociology of the coast, in trying to figure out how the sea impacts port cities and affects their dwellers. Overcoming the limitations of these theories with new ones always raises further theoretical problems and methodological doubts. Our intention was to propose a research agenda, based on the theoretical concept of urban assemblage and ANT, to support the analysis of the associations between human and non-human actors. The research problem posed is: 'How are the urban assemblages of the port cities created by action nets of natural,

technological and social actors?'. We believe that if human and non-human actor networks create the urban assemblages of port cities, the sea as the super-actor modifies their network relationships. The dominant mediator is the port (technology) that translates mutual actions of human (business, institutions, port city dwellers, etc.) and non-human actors (sea, maritime environment, etc.). The port, in social consciousness, goes beyond its dictionary understanding as it refers to the history of the port city, i.e. the development of the maritime economy, its cultural heritage and traditions created in contact with the sea over the centuries.

Nevertheless, 'the port' is still a technology that generates scenes, impacts spatial imagination, enables performances, transmits tacit knowledge, builds a city's identity and mediate associations of actants. The ensemble of multiple enactments, action nets and, eventually, actor-networks grasps the urban assemblage of the port city. If the urban assemblages and Actor-Network Theory are to change urban studies, port city research is an interesting field of research. The proposed research plan is only one possible way to capture these problems. We realise that it is not a panacea for all theoretical difficulties. Nor is it free from ontological, epistemological, and methodological problems. However, it is the research practice that should verify whether it is the proverbial "dead end".

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